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Foundations of Programming

Assignment No: 8

PROBLEM STATEMENT:

Write a program in C to accept marks of five courses of a student and compute the result. A student is considered PASS if he/she scores 40 marks or more in each course. If the student passes, calculate the aggregate percentage and assign the grade as follows:

- Aggregate ≥ 75% : Distinction
- Aggregate ≥ 60% and < 75% : First Division
- Aggregate ≥ 50% and < 60% : Second Division
- Aggregate ≥ 40% and < 50% : Third Division

OBJECTIVE:

- To understand the use of **conditional statements** in C
- To apply **logical operators** for decision making
- To calculate **total, percentage, and grade** of a student
- To enhance problem-solving skills using real-life scenarios

THEORY:

In academic evaluation systems, student performance is assessed based on individual subject marks as well as overall aggregate performance. In C programming, such evaluation systems can be implemented using conditional statements like `if`, `else if`, and `else`. Logical operators help in checking multiple conditions simultaneously. Percentage calculation involves arithmetic operations, and grading logic is implemented based on predefined ranges. This program demonstrates structured programming and decision-making in C.

ALGORITHM:

1. Start
2. Input marks of 5 subjects
3. If any mark < 40 → Print “Fail” and Stop
4. Else calculate total and percentage
5. Assign grade based on percentage
6. Print result, percentage, and grade
7. Stop

PLATFORM: Linux

PROGRAM:

```
#include <stdio.h>

// student grade evaluation

int main() {

   int marks[5], i, total = 0;

   int passed = 1;

   float percentage;

   printf("Enter marks of 5 subjects: ");

   for (i = 0; i < 5; i++) {

      scanf("%d", &marks[i]);

      total += marks[i];

      if (marks[i] < 40)

         passed = 0;

   }

   percentage = total / 5.0;

   printf("Total Marks = %d\n", total);

   printf("Percentage = %.2f%%\n", percentage);

   if (!passed) {

      printf("Result: FAIL\n");

   } else {

      printf("Result: PASS\n");

      if (percentage >= 75)
```

```
        printf("Grade: Distinction\n");
    else if (percentage >= 60)
        printf("Grade: First Division\n");
    else if (percentage >= 50)
        printf("Grade: Second Division\n");
    else
        printf("Grade: Third Division\n");
  }
  // conditional grading system
  return 0;
}
```

```c
#include <stdio.h>
int main()
{
    float m1, m2, m3, m4, m5;
    float total, percentage;
    printf("Enter marks of five courses:\n");
    scanf("%f %f %f %f %f", &m1, &m2, &m3, &m4, &m5);

    if (m1 < 40 || m2 < 40 || m3 < 40 || m4 < 40 || m5 < 40)
    {
        printf("FAIL\n");
    }
    else
    {
        total = m1 + m2 + m3 + m4 + m5;
        percentage = total / 5;

        printf("Total Marks = %.2f\n", total);
        printf("Percentage = %.2f\n", percentage);

        if (percentage >= 75)
            printf("Grade: Distinction\n");
        else if (percentage >= 60 && percentage < 75)
            printf("Grade: First Division\n");
        else if (percentage >= 50 && percentage < 60)
            printf("Grade: Second Division\n");
        else
            printf("Grade: Third Division\n");
    }
    return 0;
}
```

OUTPUT: The program displays the total marks, percentage, result, and grade.

Program executed successfully.

Example Output:Enter marks of 5 subjects: 80 85 78 90 88

Total Marks = 421

Percentage = 84.20%

Result: PASS

Grade: Distinction

CONCLUSION: Thus, the program to calculate student result and grade based on marks was successfully implemented using C programming. This program demonstrates the effective use of conditional statements, logical operators, and arithmetic operations.

FAQs:

1. What is the objective of this C program?
2. What is the objective of this C program is an important programming concept because it supports clear problem solving, structured logic, and correct implementation. Understanding it helps in writing programs that are easier to develop, test, and maintain.
3. Why is an if–else ladder used in this program?
4. Why is an if–else ladder used in this program is an important programming concept because it supports clear problem solving, structured logic, and correct implementation. Understanding it helps in writing programs that are easier to develop, test, and maintain.
5. Which data types are generally used in this program?
6. A data type specifies the kind of value a variable can store, such as integer, float, character, or string. Choosing the correct data type improves program accuracy, memory usage, and reliability.